CS-433 Machine Learning - Beyond Pixel Classification: Enforcing Topological Connectivity in Aerial Road Extraction in Limited Data Regimes

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***Abstract*—Automated extraction of road networks from satellite imagery remains challenging due to data scarcity and complex topologies. This work evaluates deep learning strategies to address these challenges using a limited dataset of 100 images. Various encoder-decoder architectures, including LinkNet and U-Net variants, are analyzed. To address data scarcity, a robust augmentation pipeline is implemented, and the impact of incorporating geometric priors through dilated convolutions and Multi-Task Learning (MTL) for road orientation prediction is investigated. Experiments demonstrate that the D-LinkNetRes34MTL architecture, which combines a pre-trained ResNet-34 encoder with joint orientation learning, achieves the highest performance with a Test F1 score of 0.917. These findings confirm that combining robust data augmentation with domain-specific topological constraints and transfer learning is essential for achieving high segmentation performance on limited datasets.**

I. Introduction

Extracting road networks from satellite imagery is vital for urban planning and navigation. Since manual annotation is too slow for large-scale mapping, deep learning offers an automated solution. This work details our approach to selecting and refining neural models for pixel-level classification.

II. Data Processing

Data Analysis: The training data consists of 100 RGB satellite images with dimensions of 400×400 pixels. Each image is paired with a binary ground-truth mask, where pixels are labeled as 'road' (white) or 'background' (black). The data has a significant class imbalance, with 77.47% pixels categorized as background and only 22.53% as road. To mitigate the bias inherent in accuracy metrics for imbalanced data, the F1-score of the road class was used to evaluate performance. The testing is conducted via the AICrowd platform on 50 images with dimensions of 608×608 pixels. Predictions are submitted as 16×16 patches, labeled 'road' if pixel density exceeds 25%.

Data Augmentation: Given the limited size of the training dataset (100 images), an extensive data augmentation pipeline was implemented to enhance diversity and mitigate algorithmic bias by forcing the model to learn shape-invariant features rather than texture-specific patterns. Augmentations are generated dynamically during training. To preserve spatial consistency, geometric transformations are applied identically to both the input image and its corresponding ground-truth mask, whereas photometric distortions are applied solely to the input image. The specific transformations and their execution probabilities (p) are:

- **Horizontal Flip ($p = 0.5$):** Randomly flips the input along the horizontal axis.
- **Vertical Flip ($p = 0.5$):** Randomly flips the input along the vertical axis.
- **Discrete Rotation ($p = 0.5$):** Rotates the input by $90°$, $180°$, or $270°$ to exploit the orientation invariance of aerial imagery.
- **Affine Transformation ($p = 0.5$):** Applies random rotation ($\pm 45°$), scaling ($\pm 10\%$), and translation ($\pm 6.25\%$) using reflection padding for boundary handling.
- **Coarse Dropout ($p = 0.2$):** Randomly masks out 3 to 6 rectangular regions with dimensions ranging from 8×8 to 15×15 pixels. This encourages the model to learn context-aware features and improves robustness against occlusion.
- **Elastic Deformation ($p = 0.2$):** Applies random, non-rigid elastic distortions to the image to simulate local topographic variations.
- **Photometric Distortion ($p = 0.5$):** Enhances invariance to lighting conditions by applying random color perturbations—brightness ($\pm 40\%$), contrast ($\pm 40\%$), and hue ($\pm 20\%$)—while leaving the ground-truth masks unaltered.

III. Baseline Models

The loss used by all the baseline models is the SoftIoU (Soft Intersection over Union) loss. SoftIoU Loss is particularly valuable because it counteracts the extreme class imbalance, where thin road networks occupy only a tiny fraction of the total aerial imagery. By optimizing for global shape overlap rather than individual pixel accuracy as done by Binary Cross Entropy loss, it prevents the model from ignoring narrow paths and encourages the prediction of continuous, connected road structures rather than disjointed segments. SoftIoU Loss is defined as:

$$L_{SoftIoU} = 1 - \frac{\sum_i (y_i \cdot \hat{y}_i) + \epsilon}{\sum_i y_i + \sum_i \hat{y}_i - \sum_i (y_i \cdot \hat{y}_i) + \epsilon}$$

where y_i is the ground truth value (0 or 1) for pixel i, $\hat{y}_i$ is the predicted probability (0 to 1) for pixel i and ϵ is a small smoothing factor (e.g., $1e-7$) to avoid division by zero.

A. Logistic Regression

As a lowest-capacity reference, a per-pixel logistic regression model was implemented for road segmentation. In the context of semantic segmentation, this corresponds to classifying each pixel independently using only its RGB values, without any spatial context. Concretely, the model can be viewed as a single 1×1 convolution that maps the 3 input channels to C output logits per pixel (with $C = 2$ for background vs. road). The prediction at pixel (i, j) is:

$$\hat{y}_{ij} = \text{softmax}(W x_{ij} + b)$$

where $x_{ij} \in \mathbb{R}^3$ is the RGB vector and $\hat{y}_{ij} \in \mathbb{R}^C$ are the class probabilities.

This baseline is useful to quantify how much of the task can be solved by simple color statistics alone, and to highlight the benefit brought by spatial feature extraction in convolutional models.

B. Plain Convolution Neural Network (Plain CNN)

Beyond logistic regression, a simple convolutional baseline was implemented to introduce spatial modeling while keeping the architecture intentionally minimal. The model is fully-convolutional and preserves the input resolution throughout the forward pass: it applies a small stack of 3×3 convolutions with Batch Normalization and ReLU nonlinearities, and produces per-pixel class logits with a final 1×1 convolution. Unlike U-Net/LinkNet-style encoder–decoder architectures, this baseline does not use pooling/upsampling or skip connections; the receptive field grows only through stacking local convolutions.

IV. Advanced Models

All advanced models utilize the SoftIoU loss, with the exception of the D-LinkNetMTL, D-LinkNetRes34MTL, and SpinD-LinkNetRes34MTL. These models employ an unweighted combination of SoftIoU and Multi-class Cross-Entropy loss, as detailed in Section IV-B2.

A. U-Net

The U-Net architecture, originally developed for biomedical image segmentation [1], contains a symmetric "U-shaped" encoder-decoder structure designed to capture context via a contracting path and precise localization via an expanding path. The encoder takes a 3-channel RGB input through four downsampling blocks. Each block applies a specialized double-convolution module which consists of two sequences of 3×3 convolution, Batch Normalization, and ReLU activation.

This module is then followed by a 2×2 Max Pooling with a stride of 2. This progressively doubles the feature channels from 32 to 256. The bottleneck section then utilizes the same double-convolution module, which expands the depth to 512. It then has a specific Dropout layer (p=0.5) which is added to mitigate overfitting.

The decoder restores the spatial resolution through four upsampling blocks. Here, unlike the original U-Net [1], the implementation uses bilinear upsampling with a scale factor of 2 as compared to the tradition approach of using transposed convolutions. The upsampled features U_l are concatenated with the corresponding high-resolution encoder maps C_{l-1} via skip connections. This is done to recover fine-grained spatial information. This combined map is then processed by the double-convolution module to reduce channels sequentially from 256 down to 32. The final segmentation map is generated by a 1×1 convolution which projects the output to the required 2 classes. This is then followed by a Softmax function.

B. LinkNet

LinkNet, originally developed for semantic segmentation for visual scenes [2], is an encoder-decoder architecture that uses a non-pretrained ResNet-34 [3] as its encoder, consisting of an initial convolutional layer followed by four encoding blocks that progressively extract features e1, e2, e3, and e4 at 64, 128, 256, and 512 channels, respectively. The decoder comprises four symmetric DecoderBlock modules that produce outputs d4, d3, d2, and d1, where each block performs channel reduction via 1×1 convolution (reducing channels by $4\times$), spatial upsampling through transposed convolution with stride 2 and kernel size 3, and channel expansion back to match skip connection dimensions, with batch normalization and ReLU activations after each operation. Skip connections are implemented by directly adding encoder features to decoder outputs at the corresponding resolution levels (e4→d4, e3→d3, e2→d2, e1→d1). The final classification head consists of a transposed convolution (64→32 channels, stride 2, kernel size 3), followed by two standard convolutions (32→32 channels with kernel size 3 and 32→2 channels with padding 1, kernel size 2), using LeakyReLU activations (negative slope 0.2) to produce the final segmentation output.

Four key enhancements were introduced to better adapt them for road segmentation.

1) Center Block Modification (D-LinkNet): In D-LinkNet [4], a Dilation Block (Dblock) is introduced in the center (bottleneck) between the encoder and the decoder. The Dblock operates on the deepest encoder feature map e4 (512 channels) and consists of four parallel dilated convolutional branches with progressively increasing dilation rates of 1, 2, 4, and 8, each using kernel size 3 with corresponding padding values (1, 2, 4, 8) to maintain spatial dimensions. Each dilated convolution is followed by ReLU activation, and the outputs from all four branches are aggregated with the input feature map through element-wise addition to capture multi-scale contextual information. This aggregated output replaces e4 before being passed to the decoder. The rest of the architecture remains identical to LinkNet including the encoder, the decoder and the final classification head.

This center block modification enables D-LinkNet to capture both local fine-grained details and global semantic context simultaneously, improving segmentation performance for complex spatial patterns.

2) Orientation Learning Strategy (D-LinkNetMTL): Building upon D-LinkNet, the D-LinkNetMTL architecture introduces a multi-task learning framework that jointly predicts road segmentation and road orientation angles to improve topological connectivity. The architecture shares the same ResNet-34 encoder and center Dblock as D-LinkNet, but branches into two parallel decoder paths after the bottleneck: a primary segmentation decoder (identical to D-LinkNet decoder) producing binary road masks (2 classes), and an auxiliary orientation decoder that replicates D-LinkNet's complete decoder structure with the sole architectural difference being the final convolutional layer (32→37 channels with padding 1, kernel size 2) which outputs 37 classes instead of 2, representing quantized angle bins. Both decoder branches process the shared encoder features e1-e4 independently with skip connections at each level, and each terminates in its own classification head.

The architecture is similar to [5] except that they use the Stacked Multi-branch architecture instead of the D-LinkNet architecture. To train the orientation branch, pixel-wise ground truth angles from the binary road masks are derived and categorized into 37 classes (36 directional bins of 10° and one background class). A detailed description of this ground-truth generation process is provided in Appendix IX-A.

Multi-class Cross-Entropy loss is used for orientation angle classification task: $\mathcal{L}_{Orient} = -\frac{1}{M}\sum_{i=1}^{M}\sum_{c=1}^{37} y_{i,c} \cdot \log(\hat{y}_{i,c})$ where M is the total number of pixels in the input image, $y_{i,c}$ is 1 if the pixel i's ground truth orientation angle belongs to orientation bin c and 0 otherwise, $\hat{y}_{i,c}$ is the predicted probability for pixel i being in bin c.

The network optimizes a joint loss of both tasks:

$$L_{total} = 1 \cdot L_{SoftIoU} + 1 \cdot L_{Orient}$$

Where $L_{SoftIoU}$ is the Soft IoU loss of road segmentation task and L_{Orient} is the Multi-class Cross-Entropy loss of orientation learning task.

3) Pretrained Backbone Initialization (D-LinkNetRes34MTL): To take advantage of transfer learning from large-scale image classification, the D-LinkNetMTL encoder is replaced with a ResNet-34 [3] pre-trained on ImageNet-1K dataset [6], while maintaining the identical architecture for the center Dblock and dual decoder branches. During training, the entire network including the pre-trained encoder is fine-tuned end-to-end. This transfer learning approach typically accelerates convergence and improves generalization, particularly when the training data is limited, as the pre-trained features provide a stronger initialization point compared to training from scratch. It uses the same loss as D-LinkNetMTL.

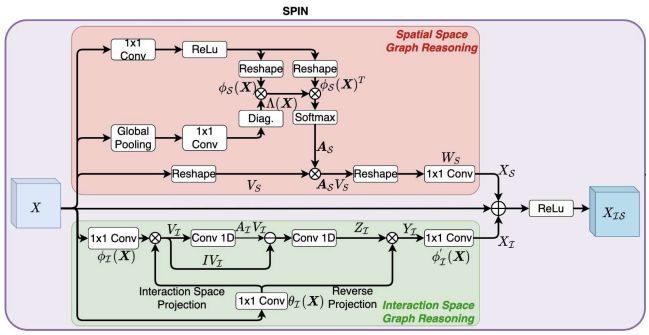


Figure 1. Spin Module Architecture. **Source: [7]**

4) Spin Attention Mechanism (SpinD-LinkNetRes34MTL): Building upon the work of [7] who developed the SPIN module to capture long-range dependencies through graph reasoning in dual spaces, the SpinD-LinkNetRes34MTL architecture integrates Spatial and Interaction Space Graph Reasoning (SPIN) modules into the D-LinkNetRes34MTL architecture. The SPIN module builds graphs in two complementary spaces: spatial space and a projected latent interaction space from input feature maps, where graph reasoning is performed in spatial space to improve connectivity between road segments and in interaction space to delineate roads from other topographies. The exact architectural details and data flow of the SPIN module are illustrated in Figure 1.

The D-LinkNetRes34MTL is modified so that a SPIN module operating on a 32-channel feature map (dgcn_seg_1 with planes=32 and ratio=1) is inserted exclusively within the final classification head of the road segmentation decoder after the initial transposed convolution and LeakyReLU activation. The orientation angle decoder branch remains unchanged. It uses the same loss as D-LinkNetMTL.

V. Results

Training was conducted on full-resolution images (400×400 pixels). To address limited data, 5-fold cross-validation was used with F1-score of the road class as the primary validation metric, averaged across the final epoch of all 5 folds. Data augmentation was applied exclusively to the training set; the validation set remained unaugmented to ensure a realistic evaluation of model generalization. After tuning, all final models were retrained on the full dataset for 200 epochs (batch size 8) using Adam (weight decay 5×10^{-4}, betas $(0.9, 0.999)$) and a Cosine Annealing Learning Rate Scheduler. All convolutional and transposed convolutional layers use Xavier normal initialization. Channel-wise mean and standard deviation were calculated from the training images. Specific preprocessing strategies and learning rates were tailored to each architecture by hyperparameter tuning; these configurations are summarized in Table II in the Appendix. Inference was performed on full-resolution images

(608×608 pixels) using a 0.5 threshold.

The specific loss functions and architectures are detailed in Section III and Section IV.

Table I shows the best results obtained from various models for comparison based on Cross-validation score and AICrowd submission Test F1 score.

Table I
MODEL F1-SCORES ON CROSS-VALIDATION AND TEST SET.

Model	CV F1 (mean ± std)	Test F1
Logistic Regression	0.3359 ± 0.0144	0.368
Plain CNN	0.6403 ± 0.0248	0.728
U-Net	0.8497 ± 0.0317	0.902
LinkNet (without Augmentation)	0.8343 ± 0.0472	0.861
LinkNet	0.8456 ± 0.0370	0.897
D-LinkNet	0.8465 ± 0.0364	0.899
D-LinkNetMTL	0.8586 ± 0.0302	0.910
D-LinkNetRes34MTL	0.8753 ± 0.0235	0.917
SpinD-LinkNetRes34MTL	0.8679 ± 0.0283	0.917

VI. DISCUSSION

Our experiments demonstrate a clear correlation between model capacity—specifically the ability to capture multi-scale spatial context—and segmentation performance. The results validate that while local color features provide a foundation, high-fidelity road extraction requires architectural mechanisms that enforce global consistency and topological connectivity.

Spatial Context and Receptive Field: The poor performance of the Logistic Regression baseline (Test F1: 0.368) confirms that pixel-wise spectral analysis is insufficient for road segmentation. Roads share similar RGB distributions with building rooftops and other concrete structures, making spatial context indispensable. While the Plain CNN improved upon this (Test F1: 0.728) by introducing local spatial features via convolutions, it lacked the downsampling operations necessary to expand the receptive field effectively. The dramatic performance leap observed with U-Net (Test F1: 0.902) and LinkNet (Test F1: 0.897) highlights the necessity of encoder-decoder architectures. By progressively downsampling to capture semantic context and upsampling to recover spatial details, these models successfully differentiate road networks from background clutter.

Impact of Data Augmentation: Given the small dataset size (100 images), overfitting was a significant risk. The comparison between LinkNet trained with and without augmentation illustrates the efficacy of our pipeline. Applying geometric and photometric transformations improved the LinkNet Test F1 score from 0.861 to 0.897. This suggests that the augmentation strategy effectively simulated a larger dataset, forcing the model to learn features invariant to rotation and lighting conditions rather than memorizing specific training examples.

Architectural Comparison: U-Net vs. LinkNet: U-Net (Test F1: 0.902) slightly outperformed LinkNet variants. This advantage likely stems from its concatenation-based skip connections, which preserve high-resolution spatial details more effectively than LinkNet's residual summation, enabling sharper delineation of thin road boundaries.

Architectural Refinements and Multi-Task Learning: The progression of LinkNet variants highlights the value of specialized modules for road topology:

- **Dilation for Global Context:** The D-LinkNet model improved upon the standard LinkNet (Test F1 0.897 vs 0.899), confirming that the central dilation block effectively expands the receptive field. This allows the network to aggregate context from a wider area without losing resolution, which is critical for maintaining the continuity of long road segments.
- **Topological Consistency via MTL:** The introduction of the orientation learning task in D-LinkNetMTL yielded a notable improvement, boosting the Test F1 to 0.910. The auxiliary loss forces the encoder to learn features sensitive to the linear and directional nature of roads. This acts as a regularizer, discouraging the prediction of disjointed blobs and encouraging connected, smooth road structures.
- **Transfer Learning:** Replacing the encoder with a ResNet-34 pre-trained on ImageNet (D-LinkNetRes34MTL) resulted in the highest performing configuration (Test F1: 0.917). Transfer learning proved particularly effective given the limited training data, as the pre-trained weights provide robust low-level feature detectors (edges, textures) that would be difficult to learn from scratch with only 100 aerial images.
- **Diminishing Returns with Complexity:** The integration of the SPIN module yielded no performance gain (Test F1: 0.917), suggesting a saturation point. This indicates that the combination of pre-training, dilation, and multi-task learning likely already maximizes information extraction from this small dataset, rendering the additional complexity of graph reasoning unnecessary.

VII. CONCLUSION

This study showcases that accurate road segmentation from limited satellite imagery requires the model to have understanding of architectural depth and geometric priors. While standard encoder-decoder models such as U-Net set a high baseline (F1 = 0.902), the best performance (F1 = 0.917) was achieved with the D-LinkNetRes34MTL architecture. Hence, this confirms that expanding the receptive field via dilated convolutions, enforcing topological consistency through multitask orientation learning and leveraging transfer learning were critical for overcoming the biggest limitation of data scarcity. However, similarly, the saturation observed with the SPIN module also suggests that incorporating domain-specific (such as road linearity) inductive biases is more effective than just increasing the model complexity.

VIII. Ethics

A significant risk of algorithmic bias was identified, which comes from the lack of diversity of the training dataset. As seen from the dataset analysis, the 100 training images have features, which are well-paved, orthogonal suburban roads with clear markings. Therefore, the model trained only on this, is likely to underperform on unpaved roads, informal settlements or complex non-grid topographies, which are often found in rural or developing regions. The stakeholders who are directly effected by this are city planner, who may receive inaccurate maps and residents of these underrepresented areas.

The negative impact of this is the potentially unequal public service delivery. If this tool is used for disaster response routing or some infrastructure development, then the "invisible" roads in these vulnerable areas could be excluded from resource allocation. This can lead to obstacles in development and planning to such underrepresented areas which need these services more as compare to other developed areas. Hence, this is assessed as high in risk, as this is highly likely to take place, due to the same and homogeneous nature of the dataset. It is also high in severity due to the potential consequences for safety and accessibility ignored in these regions.

This risk was further evaluated by analyzing the model's reliance on specific textures during the validation phase. It was observed that without data augmentation, the model overfits to the specific texture of the training data, failing to generalize to shadowed or occluded areas. To minimize this risk, an extensive data augmentation pipeline was implemented. This forced the model to learn shape-invariant features rather than relying on texture (Test F1 score improved from 0.861 to 0.897). This improved the model robustness; however, it has not completely mitigated the risk due to the limitation of the available data and also the task in hand. A truly ethical deployment would require retraining on a geographically diverse dataset representing multiple road types.

References

[1] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," 2015. [Online]. Available: https://arxiv.org/abs/1505.04597

[2] A. Chaurasia and E. Culurciello, "Linknet: Exploiting encoder representations for efficient semantic segmentation," in *2017 IEEE Visual Communications and Image Processing (VCIP)*. IEEE, Dec. 2017. [Online]. Available: http://dx.doi.org/10.1109/VCIP.2017.8305148

[3] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," 2015. [Online]. Available: https://arxiv.org/abs/1512.03385

[4] L. Zhou, C. Zhang, and M. Wu, "D-linknet: Linknet with pretrained encoder and dilated convolution for high resolution satellite imagery road extraction," in *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2018, pp. 192–1924.

[5] A. Batra, S. Singh, G. Pang, S. Basu, C. Jawahar, and M. Paluri, "Improved road connectivity by joint learning of orientation and segmentation," in *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2019.

[6] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "Imagenet: A large-scale hierarchical image database," in *2009 IEEE Conference on Computer Vision and Pattern Recognition*, 2009, pp. 248–255.

[7] W. G. C. Bandara, J. M. J. Valanarasu, and V. M. Patel, "Spin road mapper: Extracting roads from aerial images via spatial and interaction space graph reasoning for autonomous driving," in *2022 International Conference on Robotics and Automation (ICRA)*, 2022, pp. 343–350.

[8] YXDragon, "sknw: Skeleton analysis in python," 2022, analysis of object skeletons, build network from skeleton image (2D-3D). [Online]. Available: https://github.com/Image-Py/sknw

[9] D. H. Douglas and T. K. Peucker, "Algorithms for the reduction of the number of points required to represent a digitized line or its caricature," *Cartographica: The International Journal for Geographic Information and Geovisualization*, vol. 10, pp. 112–122, 1973. [Online]. Available: https://api.semanticscholar.org/CorpusID:60447873

[10] U. Ramer, "An iterative procedure for the polygonal approximation of plane curves," *Comput. Graph. Image Process.*, vol. 1, pp. 244–256, 1972. [Online]. Available: https://api.semanticscholar.org/CorpusID:1213794

IX. Appendix

A. Derivation of Ground Truth Orientation Angles

To derive pixel-wise ground truth orientation angles, ground truth images are skeletonized using the package [8] to obtain road line strings, and then smoothed using the Ramer-Douglas-Peucker algorithm [9], [10]. The coordinates of the points of each line string are sorted so that most of the directional vectors point from left to right and top to bottom. Then, unit directional vectors between consecutive annotation points of each road string are computed and converted into polar coordinates to obtain its orientation angle. Pixels falling within a 12-pixel threshold width (following [5]) along the perpendicular direction of a road string are assigned the string's orientation angle. Pixels that are not assigned any orientation angle are assigned a background class. The orientation angles are then quantized into 10° bins. This results in 37 classes, representing 36 quantized angle bins (10° intervals) and one background class.

B. Implementation Details

Table II
SUMMARY OF NORMALIZATION AND LEARNING RATES FOR EACH MODEL ARCHITECTURE.

Model	**Normalization**	**Learning Rate**
Logistic Regression	Zero-Centering	5×10^{-4}
Plain CNN	Zero-Centering	5×10^{-4}
Unet	Zero-Centering	5×10^{-4}
LinkNet (without Augmentation)	Zero-Centering	1×10^{-3}
LinkNet	Zero-Centering	1×10^{-3}
D-LinkNet	Zero-Centering	5×10^{-4}
D-LinkNetMTL	Standardization	5×10^{-4}
D-LinkNetRes34MTL	Zero-Centering	5×10^{-4}
SpinD-LinkNetRes34MTL	Standardization	1×10^{-3}